Our M1 Plan and Reproduction Results

1. Our frozen M1 plan

Research question

Can we reproduce the paper's finding that negative comments affect how urgent developers think a problem is, using the authors' original data and code?

Sample

We used data from 59 developers, who each rated the urgency of four code comments, giving us 236 ratings in total. Each developer rated two neutral comments and two negative comments.

The developers were divided into three groups based on their answer to the survey question: "Does a negative comment make a problem seem more important to you?"

- 22 agreed.
- 11 had no opinion.
- 26 disagreed.

Method

We ran the authors' original model on the urgency ratings, using a fixed random seed so the results could be reproduced.

Metric

We calculated an odds ratio for each group to compare how often negative comments were rated as more urgent rather than less urgent. An odds ratio of 1 means there is no difference.

Decision rule

Before running the experiment, we decided that the result would count as reproduced if the group that agreed with the survey statement had an odds ratio between 1.85 and 2.05, with strong evidence. The other two groups also needed to have values close to 1.

2. Our execution procedure

We ran the authors' notebook, work/analyse_urgency_seeded.Rmd, inside Docker. We used the authors' original software setup, with a few small changes to make the analysis run correctly and save the required results.

We used a single command to run the notebook and generate the output files.

3. Our raw evidence

We saved the results in the following files:

- work/analyse_urgency_seeded.html contains the full notebook output.
- work/Urgency perception-1.csv, work/Urgency perception-2.csv and work/Urgency perception-3.csv contain 100,000 simulated results for each group.
- work/evidence Urgency.csv contains the evidence strength values.

These files allow us to check the results and see how they were produced.

4. Tracing our results

We can trace the agree group's result of **1.94** back to the original data and code.

1. The result of 1.94 is recorded in RESULT.md.
2. It comes from work/Urgency perception-3.csv, where 48,741 results were classified as more urgent and 25,121 as less urgent.
3. This file was generated by running analyse_urgency_seeded.Rmd with the random seed 20261013.
4. The notebook used the authors' original data file, export_jan_23_no_ip_no_email.csv, containing ratings from 59 developers across four comments each.

This gives us a clear record of how the final result was obtained.

5. Changes made to the original setup

We made four changes, all recorded in DEVIATIONS.md before running the experiment:

1. Added a line to fix an installation problem with the authors' software setup.
2. Set a fixed random seed so the results could be reproduced.
3. Ran only the urgency analysis because it was the part relevant to our research question.
4. Saved the evidence strength values to a file because the original code only printed a general description, such as "Strong".

These changes were documented, and none changed the data, model or analysis method.

6. Current reproduction status: REPRODUCED

Our results met the criteria we established before running the experiment.

The agree group had an odds ratio of **1.94**, which falls within our target range of 1.85–2.05 and had strong evidence. The other two groups had values of 1.06 and 1.09, both close to 1.

We also checked whether the result was consistent across different runs. Three additional runs using different random seeds produced values of 1.95, 1.96 and 1.96. Saif also ran the analysis separately and obtained 1.93.

These results are very similar, which supports the conclusion that our finding is stable.

Based on our predefined decision rule, we consider the reproduction successful.

7. Held-out data

We only used the data, code and setup files originally released to our group. We did not access or use any Event or held-out material.

This means we stayed within the limits of our original M1 plan.